**Newton School of Technology, Rishihood University**

Machine Learning

# Traffic Speed Forecasting: AML Baselines and Graph Learning

Project Report

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**Project GitHub Repository**

github.com/ArPriCode/intelligent-traffic-forecasting

# Declaration

We declare that this project report, titled *Traffic Speed Forecasting: AML Baselines and Graph Learning*, documents the work carried out by Team Quanta as part of the Applied Machine Learning course at Newton School of Technology, Rishihood University.

The report describes the datasets, preprocessing procedures, forecasting models, evaluation protocol, experimental results, and limitations of the implemented system. Reported numerical results are based on the project outputs supplied for this study. Any limitations or unverified claims are identified explicitly.

**Team Quanta**

**Institution:** Newton School of Technology, Rishihood University

# Acknowledgements

We would like to thank our course instructors and the university for providing the opportunity to undertake this Machine Learning project. We also acknowledge the researchers who released the METR-LA traffic dataset and the DCRNN implementation, which provides useful sensor metadata and graph-construction references for traffic forecasting.

Finally, we thank all members of Team Quanta for their contributions to data analysis, model development, experimentation, and documentation.

# Abstract

Accurate traffic-speed forecasting is an important component of intelligent transportation systems. This project compares four forecasting approaches on five-minute traffic-speed observations from the METR-LA dataset: persistence, a segment-wise multi-output Random Forest, pooled XGBoost, and a trained Graph Attention Network (GAT).

The evaluation follows chronological training, validation, and test partitions of 70%, 10%, and 20%, respectively. Forecast horizons are 5, 15, 30, and 60 minutes. Missing and zero-speed observations are handled using causal input filling and masked target evaluation. Performance is assessed using Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Squared Error (RMSE).

The experiments show that Random Forest achieves the lowest overall MAE at all four horizons. The GAT incorporates spatial attention over a fixed graph of 207 sensors, but it does not outperform the tree-based models in the reported run. This result highlights that incorporating graph structure does not automatically improve forecasting accuracy.

The project also examines peak and off-peak performance, graph construction, data quality, and limitations of the experimental setup. The available traffic CSVs contain speed observations only; therefore, flow and occupancy are not treated as measured variables. The principal contribution is a reproducible comparison of conventional machine learning baselines and graph-based forecasting under a chronological evaluation protocol.

**Keywords:** Traffic forecasting, METR-LA, Random Forest, XGBoost, Graph Attention Network, spatial graph, time-series prediction.

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# Chapter 1

# Introduction

## 1.1 Background

Traffic congestion creates delays, increases fuel consumption, and reduces the efficiency of urban transportation networks. Traffic-speed forecasting attempts to estimate future traffic conditions using historical observations. These predictions can support traffic monitoring, route planning, and transportation decision-making.

Traffic measurements are spatially related: conditions at one road segment can be associated with conditions at nearby segments. A useful forecasting system should therefore consider both temporal patterns and relationships among traffic sensors.

Traditional machine learning methods can learn temporal patterns from engineered features. Graph-based models provide another approach by representing sensors as nodes and their relationships as edges.

## 1.2 Problem Statement

The objective of this project is to compare conventional machine learning baselines with a Graph Attention Network for forecasting traffic speed at multiple future horizons using historical METR-LA observations.

The project investigates the following questions:

1. How accurately can persistence, Random Forest, and XGBoost forecast traffic speed?
2. Does graph attention improve forecasting performance over the tree-based baselines?

3. How does forecasting error change as the prediction horizon increases?
4. How do peak and off-peak traffic conditions affect model performance?

## 1.3 Objectives

The main objectives are:

- Examine the traffic datasets and their data-quality issues.
- Establish a persistence baseline.
- Train segment-wise Random Forest and pooled XGBoost models.
- Construct a spatial graph for the METR-LA sensors.
- Train a Graph Attention Network using historical speed data.
- Evaluate all models on a chronological test partition.
- Compare forecasting accuracy across horizons and traffic periods.
- Document limitations and identify directions for future work.

## 1.4 Scope

The primary evaluation focuses on traffic-speed forecasting using METR-LA. PEMS-BAY is included in the dataset and exploratory-analysis documentation, but the quantitative model-comparison results presented in this report are for METR-LA.

The project does not claim that its forecasts constitute a deployed traffic-control system. A signal-control simulation result is included only as an optional experiment and is distinguished from the validated forecasting benchmark.

# Chapter 2

# Literature Review

## 2.1 Traffic-Speed Forecasting

Traffic forecasting has been approached using statistical time-series models, conventional machine learning, and deep learning. Historical speed observations provide useful information about future speed, but prediction becomes more challenging as the forecast horizon increases.

A persistence baseline is particularly useful because it measures the accuracy obtained by assuming that the most recently observed speed will remain unchanged.

## 2.2 Tree-Based Machine Learning

Random Forest combines predictions from an ensemble of decision trees. It can model nonlinear relationships among historical speed values, time features, and sensor-specific patterns.

XGBoost uses gradient-boosted decision trees and can learn complex relationships through sequential optimization. In this project, a pooled sensor-aware XGBoost model is used to learn across multiple sensors.

## 2.3 Graph-Based Traffic Forecasting

Graph-based approaches represent a sensor network as a graph $G = (V, E)$, where $V$ denotes sensors and $E$ denotes relationships between sensors. This representation allows a model to aggregate information from neighboring nodes.

Graph Attention Networks learn attention weights over graph neighborhoods, allowing the model to assign different importance to different connected sensors.

The GAT architecture was introduced by Veličković et al. [**?**]. Diffusion Convolutional Recurrent Neural Networks provide another influential graph-based approach to traffic forecasting [**?**].

## 2.4 Research Motivation

The project investigates whether spatial attention improves forecasting accuracy over carefully evaluated conventional baselines. Its results also demonstrate the importance of reporting negative results: a more complex model should not automatically be considered superior when its held-out metrics are worse.

# Chapter 3

# Dataset Description and Exploratory Data Analysis

## 3.1 Datasets

The project examines two public traffic-speed datasets: METR-LA and PEMS-BAY. The primary model-comparison experiments reported in this document use METR-LA.

Table 3.1: Dataset overview

| Dataset | Time steps | Sensors | Date range | Missing/zero |
|---|---|---|---|---|
| METR-LA | 34,272 | 207 | 2012-03-01 to 2012-06-27 | 8.11% |
| PEMS-BAY | 52,116 | 325 | 2017-01-01 to 2017-06-30 | 0.0031% |

The datasets contain speed observations at five-minute intervals. The reported missing/zero percentages are dataset-level summaries provided by the project analysis.

## 3.2 Data Quality

Zero speeds are treated as missing observations in the forecasting pipeline. This decision prevents zero-coded missing readings from being interpreted as genuine stationary traffic measurements.

Forward filling is used only to construct causal model inputs. Missing targets are excluded from target training and evaluation metrics. This prevents future observations from being used to fill earlier inputs and prevents unavailable targets from contributing to the reported errors.

The supplied traffic CSVs contain speed only. Flow and occupancy are not available as measured variables and are not imputed as though they were observed.

## 3.3 Exploratory Data Analysis

The project exploratory analysis includes:

- Traffic-speed time-series plots.
- Hourly and weekday patterns.
- Sensor-level heatmaps and speed distributions.
- Missingness and zero-speed analysis.
- Sensor correlations.
- Sensor-coordinate maps and road metadata where available.

These analyses help identify temporal patterns, data-quality issues, and spatial variation among sensors.

## 3.4 METR-LA Sensor Graph

The METR-LA graph uses official sensor IDs, locations, and road distances sourced from the DCRNN repository at commit `602afd9d767d3aa1c9b3eac51710d6aeee12c227`.

The official ID order aligns with all 207 METR-LA traffic columns. A Gaussian distance kernel and threshold of 0.1 are used to construct the distance-weighted adjacency. The resulting matrix is symmetrized, and self-loops are added for graph learning.

Table 3.2: METR-LA graph characteristics

| **Characteristic** | **Value** |
|---|---|
| Number of nodes | 207 |
| Undirected links | 1,313 |
| Mean degree | Approximately 12.7 |
| Graph type | Symmetrized distance graph |
| Self-loops | Added for model computation |

The graph represents spatial proximity based on road distance. It does not encode verified upstream/downstream direction, and it should not be interpreted as a fully directed traffic-flow network.

# Chapter 4

# Data Preprocessing and Feature Engineering

## 4.1 Chronological Splitting

The time axis is divided chronologically into training, validation, and test subsets.

Table 4.1: Chronological data partition

| **Partition** | **Proportion** |
|---|---|
| Training | 70% |
| Validation | 10% |
| Testing | 20% |

Chronological splitting preserves the direction of time and avoids randomly mixing future observations into the training partition.

## 4.2 Forecast Horizons

The models forecast traffic speed at four horizons.

Table 4.2: Forecast horizons

| Horizon | Number of five-minute intervals |
|---|---|
| 5 minutes | 1 |
| 15 minutes | 3 |
| 30 minutes | 6 |
| 60 minutes | 12 |

## 4.3 Input Processing

The input pipeline applies the following procedures:

1. Preserve the chronological order of the observations.
2. Treat zero-speed readings as missing.
3. Forward-fill missing readings only for causal inputs.
4. Exclude missing target readings from training and evaluation.
5. Fit sensor-specific scaling parameters using training timestamps only.
6. Construct historical speed lags and temporal features.

The GAT uses six consecutive five-minute speed observations, equivalent to 30 minutes of input history. The tree models use causal speed lags and time features, with no more than 25 minutes of input history as specified by the project evaluation protocol.

## 4.4 Temporal Features

Cyclic time features represent the periodic nature of time of day. For time-of-day angle $\theta_t$, sine and cosine features can be written as

$$x_{\text{hour},1} = \sin(\theta_t), \qquad x_{\text{hour},2} = \cos(\theta_t).$$

Equivalent cyclic representations can be used for weekday patterns. These features allow the models to learn recurring temporal patterns without treating the beginning and end of a cycle as unrelated values.

## 4.5 Target Masking

Let $y_{i,t}$ be the observed speed for sensor $i$ at time $t$. Define an observation mask

$$m_{i,t} = \begin{cases} 1, & \text{if the target is observed and nonzero,} \\ 0, & \text{otherwise.} \end{cases}$$

Only observations with $m_{i,t} = 1$ contribute to target losses and evaluation metrics. For the GAT, the training objective uses a masked loss so that unavailable target values do not contribute to the optimization.

# Chapter 5

# Methodology and System Architecture

## 5.1 Pipeline Overview

The system follows a sequential forecasting pipeline:

1. Load the traffic-speed observations and sensor metadata.
2. Perform exploratory data analysis and data-quality checks.
3. Construct chronological training, validation, and test sets.
4. Generate causal speed and temporal features.
5. Train the persistence, Random Forest, XGBoost, and GAT models.
6. Evaluate the models at the four forecast horizons.
7. Compare overall and period-specific forecasting errors.

## 5.2 Persistence Baseline

Persistence predicts that future speed equals the latest available observed speed:

$$\hat{y}_{t+h} = y_t.$$

This baseline is simple but important. It measures how well a model must perform to improve on the assumption that the latest observed condition continues into the future.

## 5.3 Random Forest

A multi-output Random Forest model is trained separately for each sensor. The model uses causal speed lags and cyclic time features to predict the four forecast horizons.

The training procedure uses up to 2,000 training timestamps per sensor. The segment-wise design allows each model to learn sensor-specific behavior.

## 5.4 Pooled XGBoost

Pooled XGBoost combines observations from multiple sensors into sensor-aware training data. Its feature family includes speed lags, cyclic time features, and sensor identity.

Up to 50,000 observed training targets are used per horizon. Pooling allows the model to learn shared patterns while retaining information about which sensor produced each observation.

## 5.5 Graph Attention Network

The GAT learns representations from traffic speeds and neighboring sensors in the fixed graph. Attention allows the model to weight connected sensor information rather than treating every neighbor as equally important.

The implementation uses six consecutive five-minute speed readings and cyclic time/weekday features. It uses per-sensor scaling fitted only on training timestamps, masked loss, and validation-based early stopping.

The model was trained on Apple Metal Performance Shaders (MPS) for six epochs. The best reported validation masked MSE was 0.4220.

## 5.6 Fairness of Comparison

All four models use the same test origins and valid target counts for each horizon. However, the feature representations and training volumes differ. Random Forest is capped at 2,000 training timestamps per sensor, XGBoost at 50,000 observed targets per horizon, and GAT uses eligible training windows.

Consequently, the experiment is a controlled chronological comparison, but it is not an equal-compute benchmark. The results should be interpreted in light of these differences.

# Chapter 6

# Evaluation Metrics and Experimental Protocol

## 6.1 Mean Absolute Error

Mean Absolute Error measures the average absolute difference between observed and predicted speed:

$$\text{MAE} = \frac{1}{N} \sum_{j=1}^{N} |y_j - \hat{y}_j|.$$

Lower MAE indicates smaller average absolute prediction errors.

## 6.2 Mean Absolute Percentage Error

Mean Absolute Percentage Error is calculated as

$$\text{MAPE} = \frac{100}{N} \sum_{j=1}^{N} \left| \frac{y_j - \hat{y}_j}{y_j} \right|.$$

Only observed nonzero targets are included, preventing division by zero for excluded observations.

## 6.3 Root Mean Squared Error

Root Mean Squared Error is defined as

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{j=1}^{N}(y_j - \hat{y}_j)^2}.$$

RMSE penalizes larger errors more strongly than MAE.

## 6.4 Peak and Off-Peak Definitions

Peak traffic periods are defined as weekdays from 07:00 to 09:59 and 16:00 to 18:59. All other times, including weekends, are classified as off-peak.

The evaluation reports MAE, MAPE, and RMSE over observed nonzero targets. The same valid target counts are used across the models at each horizon, enabling direct comparison of the reported metrics.

# Chapter 7

# Experimental Results

## 7.1 Overall Held-Out Results

Table 7.1 presents the overall METR-LA test results for all four forecasting approaches.

Table 7.1: Overall held-out METR-LA metrics

| Horizon | Model | MAE | MAPE (%) | RMSE |
|---|---|---|---|---|
| 5 min | Persistence | 2.814 | 6.489 | 4.627 |
| 5 min | Random Forest | **2.519** | 6.327 | **4.281** |
| 5 min | XGBoost | 2.554 | 6.435 | 4.326 |
| 5 min | GAT | 2.879 | 7.206 | 4.511 |
| 15 min | Persistence | 3.499 | 8.612 | 6.418 |
| 15 min | Random Forest | **3.119** | **8.431** | **5.702** |
| 15 min | XGBoost | 3.239 | 8.846 | 5.873 |
| 15 min | GAT | 3.466 | 9.168 | 5.830 |
| 30 min | Persistence | 4.207 | 10.858 | 8.052 |
| 30 min | Random Forest | **3.635** | **10.401** | **6.706** |
| 30 min | XGBoost | 3.854 | 11.106 | 6.988 |
| 30 min | GAT | 4.087 | 11.434 | 6.965 |
| 60 min | Persistence | 5.381 | 14.535 | 10.345 |
| 60 min | Random Forest | **4.192** | **12.623** | **7.657** |
| 60 min | XGBoost | 4.542 | 13.696 | 7.982 |
| 60 min | GAT | 4.898 | 14.328 | 8.227 |

*Note:* Lower values indicate better performance for all three metrics. Bold values indicate the best result within each horizon and metric.

## 7.2 Valid Target Counts

The number of valid target observations decreases slightly with the forecast horizon because the evaluation requires valid target readings at the corresponding future timestamps.

Table 7.2: Valid held-out target counts

| **Horizon** | **Valid targets per model** |
|---|---|
| 5 minutes | 1,244,846 |
| 15 minutes | 1,244,834 |
| 30 minutes | 1,244,816 |
| 60 minutes | 1,244,780 |

All models share the same valid target count within each horizon.

## 7.3 Overall MAE Comparison

Figure placeholder: add the model-comparison plot from `results/figures/27_metr_la_peak_offpeak_model_comparison.png`.

Figure 7.1: Peak and off-peak forecasting comparison.

The results show that Random Forest achieves the lowest overall MAE at every forecast horizon. Its MAE increases from 2.519 at five minutes to 4.192 at 60 minutes, illustrating the greater difficulty of predicting traffic speed farther into the future.

XGBoost also improves upon persistence in overall MAE at every horizon, but it remains behind Random Forest in this run.

The GAT has higher overall MAE than both tree-based approaches at all four horizons. It improves upon persistence at 15, 30, and 60 minutes, but not at five minutes. Thus, the graph-based architecture is implemented and trained, but it does not establish an accuracy advantage in these experiments.

## 7.4 Peak-Period MAE

Table 7.3: Peak-period MAE

| Model | 5 min | 15 min | 30 min | 60 min |
|---|---|---|---|---|
| Persistence | 3.287 | 4.565 | 5.831 | 7.804 |
| Random Forest | **3.189** | **4.341** | **5.324** | **6.374** |
| XGBoost | 3.243 | 4.535 | 5.808 | 7.277 |
| GAT | 3.601 | 4.737 | 6.001 | 7.940 |

Random Forest has the lowest peak-period MAE at every horizon. Errors are higher during peak periods than during off-peak periods for all four models, consistent with peak traffic being more difficult to forecast in the evaluated data.

## 7.5 Off-Peak MAE

Table 7.4: Off-peak MAE

| Model | 5 min | 15 min | 30 min | 60 min |
|---|---|---|---|---|
| Persistence | 2.695 | 3.230 | 3.797 | 4.770 |
| Random Forest | **2.350** | **2.812** | **3.210** | **3.643** |
| XGBoost | 2.381 | 2.913 | 3.362 | 3.853 |
| GAT | 2.696 | 3.146 | 3.605 | 4.131 |

Random Forest also achieves the lowest off-peak MAE at all four horizons. The gap between peak and off-peak MAE indicates that traffic conditions vary in predictability across time periods.

## 7.6 Discussion

Three major observations emerge from the results.

1. **Random Forest is the strongest evaluated model.** It achieves the lowest overall MAE at every forecast horizon and also leads the peak and off-peak comparisons.

2. **Forecasting becomes more difficult at longer horizons.** All four models show increasing MAE as the forecast horizon increases from five to 60 minutes.

3. **Graph attention does not guarantee better accuracy.** Although the GAT uses a spatial sensor graph and learned attention, it does not outperform the tree-based baselines in the reported run.

These findings suggest that conventional tree-based models remain strong baselines for the particular data preparation, features, training budgets, and evaluation protocol used in this project.

# Chapter 8

# Optional Signal-Control Simulation

## 8.1 Scope and Status

An optional experiment is described in the project materials as an intersection queuing simulation implemented in `notebooks/09_signal_simulation.ipynb`. The experiment is separate from the validated traffic-speed forecasting benchmark.

The project materials describe an experiment comparing a fixed-time signal policy with a forecast-aware adaptive policy. They report a 42.1% reduction in total intersection vehicle delay during peak congestion periods.

**Important qualification:** this percentage should be treated as a reported simulation result, not as a measured real-world traffic improvement. It should be retained in a submitted report only after the simulation notebook, generated figure, demand assumptions, and delay calculations have been checked and the result reproduced.

## 8.2 Reported Policies

The project description identifies the following policies:

- **Fixed-time policy:** a static 90-second signal cycle with a reported 53%/36% green-time split.
- **Forecast-aware adaptive policy:** dynamically adjusts green splits according to anticipated queue buildup using a max-pressure approach.

The simulation description also refers to the Greenshields fundamental diagram,

$$q = kv,$$

where $q$ denotes traffic flow, $k$ denotes density, and $v$ denotes speed. Since the supplied traffic CSVs contain speed only, any inferred demand or density used in the simulation must be documented as a modeling assumption rather than a directly measured dataset variable.

## 8.3 Interpretation

If reproduced successfully, the simulation can serve as an exploratory demonstration of how forecasts might inform signal timing under specified assumptions. It does not establish that the forecasting model controls actual road infrastructure or that the reported delay reduction would generalize to real intersections.

Optional figure placeholder: add the verified simulation output from `results/figures/28_simulated_signal_policy_experiment.png`.

Figure 8.1: Optional simulated signal-policy experiment. Include the figure only after verifying the underlying simulation results.

# Chapter 9

# Limitations

## 9.1 Data Limitations

The traffic CSVs provide speed observations only. Flow and occupancy are unavailable as directly measured variables. The report therefore does not treat them as observed input features.

Zero-speed values are treated as missing according to the project pipeline. This is a preprocessing assumption and may not distinguish every genuine zero-speed event from a missing reading.

## 9.2 Graph Limitations

The graph is constructed from road distances and symmetrized. Although this provides a spatial representation of the sensor network, it does not encode verified upstream/downstream direction or direct travel times between sensors.

## 9.3 Model Comparison Limitations

The models use different feature representations and training volumes. Random Forest is capped at 2,000 training timestamps per sensor, XGBoost at 50,000 observed targets per horizon, and GAT uses eligible training windows. This means the comparison does not equalize computational cost, number of training examples, or model capacity.

The GAT was trained for six epochs on MPS, with a best validation masked MSE of 0.4220. Further training, tuning, alternative graph construction, or architectural changes could change its results.

## 9.4 Signal-Control Limitations

The forecasting experiment does not constitute live traffic control. Any signal-policy simulation is a separate optional deliverable and requires independent verification.

Real-world signal-control evaluation would require calibrated traffic demand, intersection geometry, phase sequences, detector actuation data, and a suitable microscopic traffic simulator such as SUMO or CityFlow. The available project results do not establish deployment readiness.

# Chapter 10

# Conclusion and Future Work

## 10.1 Conclusion

This project evaluated persistence, segment-wise Random Forest, pooled XGBoost, and Graph Attention Network models for traffic-speed forecasting on METR-LA at 5-, 15-, 30-, and 60-minute horizons.

Random Forest achieved the lowest overall MAE at every horizon: 2.519, 3.119, 3.635, and 4.192, respectively. It also achieved the lowest MAE in both peak and off-peak evaluations. XGBoost performed better than persistence in overall MAE but did not surpass Random Forest.

The GAT implemented spatial attention over a fixed 207-node sensor graph. However, its overall MAE was higher than that of both tree models across all evaluated horizons. This is a useful negative result: the added complexity of graph learning did not translate into better forecast accuracy in this experimental run.

The project also documented data quality, causal preprocessing, chronological evaluation, and limitations in graph directionality and training-budget comparability. Any signal-control simulation remains a separate exploratory experiment and is not evidence of real-world traffic-control performance.

## 10.2 Future Work

Possible extensions include:

- Tune GAT depth, attention heads, hidden dimensions, dropout, learning rate, and training duration.

- Compare the fixed distance graph with learned or directed sensor graphs.
- Conduct multiple-seed experiments and report variability.
- Compare models under more closely matched training and computational budgets.
- Evaluate additional metrics and inspect per-sensor errors.
- Validate results on PEMS-BAY using the same evaluation protocol.
- If signal control remains in scope, independently validate the simulation with documented demand assumptions and calibrated traffic-simulation scenarios.

# Reproducibility and Project Deliverables

The project repository is available at:

https://github.com/ArPriCode/intelligent-traffic-forecasting

The reported project materials identify the following notebooks:

- `notebooks/05_baselines.ipynb`: baseline models.
- `notebooks/06_xgboost.ipynb`: pooled XGBoost.
- `notebooks/07_backtest.ipynb`: chronological evaluation.
- `notebooks/08_gat_model.ipynb`: graph construction and GAT.
- `notebooks/09_signal_simulation.ipynb`: optional signal simulation.

The project materials also identify metrics, predictions, adjacency data, model state, and figures under the `results/metrics/`, `results/predictions/`, and `results/figures/` directories.

The detailed metrics are reported as available in `results/metrics/metr_la_model_comparison.csv`. Reproduction should use the repository's actual files and recorded configuration to verify all numerical results.